

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO1 – Imitation (L1)	Assessment:	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA		
CO1 Statement:	Analyse NLP models, regular expression patterns, finite state automata, morphological processing techniques and to interpret language processing. (BL-3)		
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Section / Batch:	CSE AI	Date:	16\7\2026

Code of Conduct

I A. MOUNISH / 192472273 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- Develop a modular and efficient Python program that satisfies all the functional requirements specified in the problem statement.
- Demonstrate the correctness of the implementation using suitable test cases and justify the output obtained.
- Follow good programming practices, including meaningful variable names, proper code indentation, comments, and error handling wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
Information Extraction	Regex-Based Information Extraction	1
Pattern Matching	Regex Pattern Matching	2
Regular Expression Patterns	Regex-Based Input Validation	3

Section 1: Regular Expressions – Resume Information Extraction

1. A multinational recruitment company receives thousands of resumes every day in text format. The HR department requires an automated system to extract candidate information for shortlisting.

Design and implement a Python-based resume information extraction system using Regular Expressions that performs the following tasks:

Extract the candidate's name.

Identify email addresses.

Extract mobile numbers.

Detect technical skills (Python, Java, SQL, Machine Learning, NLP).

Extract years of experience.

Generate a structured summary of the candidate's profile.

Display candidates who satisfy the minimum eligibility criteria (minimum 2 years of experience and Python skill).

Section 2: Regular Expressions – Product Search System

2. An e-commerce company plans to enhance its product search engine to improve customer experience through flexible keyword matching.

Design and implement a Python-based product search system using Regular Expressions that performs the following tasks:

Search products using an exact keyword.

Perform prefix-based searches.

Perform suffix-based searches.

Search using partial keywords.

Perform case-insensitive searches.

Display all matching product names.

Generate a report showing the total number of matching products for each search.

Section 3: Regular Expressions – University Registration System

A university is developing an automated online registration portal to verify student information before course enrollment.

Design and implement a Python-based validation system using Regular Expressions that performs the following tasks:

Validate the student register number.

Validate the institutional email address.

Validate the course code.

Validate the semester information.

Validate the student's mobile number.

Display appropriate validation messages for each field.

Generate a final registration status report indicating whether the registration is successful.

1. Experiment: Resume Information Extraction Using Regular Expressions in Python

AIM:

To design and implement a Python-based Resume Information Extraction System using Regular Expressions to automatically extract candidate details such as name, email, mobile number, skills, and experience, and shortlist eligible candidates.

ALGORITHM:

1. Start the program.
 2. Import the re module.
 3. Read the resume text.
 4. Extract the candidate name using Regular Expression.
 5. Extract email address using a suitable pattern.
 6. Extract mobile number using a suitable pattern.
 7. Search for technical skills (Python, Java, SQL, Machine Learning, NLP).
 8. Extract years of experience from the resume.
 9. Generate a structured summary of the candidate profile.
 10. Check eligibility:
 - Experience $\geq$ 2 years
 - Python skill available
 11. Display candidate details and eligibility status.
 12. Stop the program.
-

PYTHON PROGRAM

```
import
```

```
# Sample Resume Text
```

```
resume = """
```

```
Name: John Smith
```

```
Email: johnsmith@gmail.com
```

```
Mobile: 9876543210
```

```
Skills: Python, SQL, Machine Learning, NLP
```

```
Experience: 3 years
```

```
"""
```

```
# Extract Name
```

```
name = re.search(r"Name:\s*([A-Za-z ]+)", resume)
```

```
name = name.group(1) if name else "Not Found"
```

```
# Extract Email
```

```
email = re.search(r'[\w\.-]+@[\w\.-]+\.\w+', resume)
```

```
email = email.group() if email else "Not Found"
```

```
# Extract Mobile Number
```

```
mobile = re.search(r'\b\d{10}\b', resume)
```

```
mobile = mobile.group() if mobile else "Not Found"
```

```
# Extract Skills
```

```
skill_list = ["Python", "Java", "SQL", "Machine Learning", "NLP"]
```

```
skills_found = []
```

```
for skill in skill_list:
```

```
    if re.search(skill, resume, re.IGNORECASE):
```

```
        skills_found.append(skill)
```

```
# Extract Experience
```

```
exp = re.search(r'(\d+)\s*years?', resume, re.IGNORECASE)
```

```
experience = int(exp.group(1)) if exp else 0
```

```
# Generate Summary
```

```
print("----- Candidate Profile Summary -----")
```

```
print("Name:", name)
```

```
print("Email:", email)
```

```
print("Mobile:", mobile)
```

```
print("Skills:", ", ".join(skills_found))
```

```
print("Experience:", experience, "Years")
```

```
# Eligibility Check
```

```
if experience >= 2 and "Python" in skills_found:
```

```
    print("\nStatus: Eligible for Shortlisting")
```

```
else:
```

```
    print("\nStatus: Not Eligible")
```

SAMPLE INPUT

Name: John Smith

Email: johnsmith@gmail.com

Mobile: 9876543210

Skills: Python, SQL, Machine Learning, NLP

Experience: 3 years

SAMPLE OUTPUT

----- Candidate Profile Summary -----

Name: John Smith

Email: johnsmith@gmail.com

Mobile: 9876543210

Skills: Python, SQL, Machine Learning, NLP

Experience: 3 Years

Status: Eligible for Shortlisting

RESULT

Thus, the Resume Information Extraction System was successfully implemented using Python Regular Expressions to extract candidate details and shortlist eligible candidates based on experience and Python skill requirements.

2. Experiment: Product Search System Using Regular Expressions in Python

AIM

To design and implement a Python-based Product Search System using Regular Expressions for exact, prefix, suffix, partial, and case-insensitive keyword matching.

\

ALGORITHM

1. Start the program.
 2. Import the re module.
 3. Create a list of product names.
 4. Accept a search keyword.
 5. Perform:
 - Exact keyword search.
 - Prefix-based search.
 - Suffix-based search.
 - Partial keyword search.
 - Case-insensitive search.
 6. Store all matching products.
 7. Display the matching product names.
 8. Count the number of matches for each search type.
 9. Generate and display a search report.
 10. Stop the program.
-

PYTHON PROGRAM

```
import re
```

```
# Product List
```

```
products = [  
    "Laptop",  
    "Laptop Bag",  
    "Gaming Laptop",  
    "Smartphone",  
    "Phone Case",  
    "Bluetooth Speaker",  
    "Wireless Mouse",  
    "Mouse Pad",  
    "Smart Watch",  
    "Watch Charger"  
]
```

```
keyword = "Laptop"
```

```
# Exact Search
```

```
exact_match = [p for p in products if re.fullmatch(keyword, p, re.IGNORECASE)]
```

```
# Prefix Search
```

```
prefix_match = [p for p in products if re.search(r'^' + keyword, p, re.IGNORECASE)]
```

```
# Suffix Search
```

```
suffix_match = [p for p in products if re.search(keyword + r'$', p, re.IGNORECASE)]
```

```
# Partial Search
```

```
partial_match = [p for p in products if re.search(keyword, p, re.IGNORECASE)]
```

```
# Case-Insensitive Search
```

```
case_insensitive_match = [p for p in products if re.search(keyword, p, re.IGNORECASE)]
```

```
# Display Results
```

```
print("EXACT SEARCH:")
```

```
print(exact_match)
```

```
print("\nPREFIX SEARCH:")
```

```
print(prefix_match)
```

```
print("\nSUFFIX SEARCH:")
```

```
print(suffix_match)

print("\nPARTIAL SEARCH:")
print(partial_match)

print("\nCASE-INSENSITIVE SEARCH:")
print(case_insensitive_match)

# Report
print("\n----- SEARCH REPORT -----")
print("Exact Matches:", len(exact_match))
print("Prefix Matches:", len(prefix_match))
print("Suffix Matches:", len(suffix_match))
print("Partial Matches:", len(partial_match))
print("Case-Insensitive Matches:", len(case_insensitive_match))
```

SAMPLE INPUT

Keyword = Laptop

SAMPLE OUTPUT

EXACT SEARCH:

['Laptop']

PREFIX SEARCH:

['Laptop', 'Laptop Bag']

SUFFIX SEARCH:

['Laptop', 'Gaming Laptop']

PARTIAL SEARCH:

['Laptop', 'Laptop Bag', 'Gaming Laptop']

CASE-INSENSITIVE SEARCH:

['Laptop', 'Laptop Bag', 'Gaming Laptop']

----- SEARCH REPORT -----

Exact Matches: 1

Prefix Matches: 2

Suffix Matches: 2

Partial Matches: 3

Case-Insensitive Matches: 3

RESULT

Thus, the Product Search System was successfully implemented using Python Regular Expressions to perform exact, prefix, suffix, partial, and case-insensitive searches, and the matching products along with the search report were displayed successfully.

3. Experiment: Student Registration Validation System Using Regular Expressions

AIM

To design and implement a Python-based Student Registration Validation System using Regular Expressions to validate student details before course enrollment.

ALGORITHM

1. Start the program.
 2. Import the re module.
 3. Read the student register number.
 4. Validate the register number using a Regular Expression.
 5. Read the institutional email address.
 6. Validate the email format using a Regular Expression.
 7. Read the course code.
 8. Validate the course code using a Regular Expression.
 9. Read the semester information.
 10. Validate the semester value using a Regular Expression.
 11. Read the mobile number.
 12. Validate the mobile number using a Regular Expression.
 13. Display validation messages for each field.
 14. If all fields are valid, display "Registration Successful".
 15. Otherwise, display "Registration Failed".
 16. Stop the program.
-

PYTHON PROGRAM

```
import re
```

```
# Sample Student Details
```

```
register_no = "23CS101"
```

```
email = "student@university.edu"
```

```
course_code = "CS501"
```

```
semester = "5"
```

```
mobile = "9876543210"
```

```
status = True
```

```
# Register Number Validation
```

```

if re.fullmatch(r'\d{2}[A-Z]{2}\d{3}', register_no):
    print("Register Number: Valid")
else:
    print("Register Number: Invalid")
    status = False

# Email Validation
if re.fullmatch(r'[a-zA-Z0-9._%+-]+@university\.edu', email):
    print("Email Address: Valid")
else:
    print("Email Address: Invalid")
    status = False

# Course Code Validation
if re.fullmatch(r'[A-Z]{2}\d{3}', course_code):
    print("Course Code: Valid")
else:
    print("Course Code: Invalid")
    status = False

# Semester Validation
if re.fullmatch(r'[1-8]', semester):
    print("Semester: Valid")
else:
    print("Semester: Invalid")
    status = False

# Mobile Number Validation
if re.fullmatch(r'[6-9]\d{9}', mobile):
    print("Mobile Number: Valid")
else:
    print("Mobile Number: Invalid")
    status = False

# Final Registration Status
print("\n----- REGISTRATION STATUS REPORT -----")

if status:
    print("Registration Successful")
else:
    print("Registration Failed")

```

SAMPLE INPUT

Register Number : 23CS101

Email Address : student@university.edu

Course Code : CS501

Semester : 5

Mobile Number : 9876543210

SAMPLE OUTPUT

Register Number: Valid

Email Address: Valid

Course Code: Valid

Semester: Valid

Mobile Number: Valid

----- REGISTRATION STATUS REPORT -----

Registration Successful

RESULT

Thus, the Student Registration Validation System was successfully implemented using Python Regular Expressions to validate the register number, institutional email address, course code, semester information, and mobile number, and generate the final registration status report.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Q. No. / Criteria	Understanding of Problem Context	Application of Concepts	Problem-Solving Approach	Accuracy of Solution	Clarity	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director

Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____